

Uma Gunturi

Master of Science in Computer Science | Virginia Tech

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Education

May 2023 Aug 2021	Virginia Tech Department of Computer Science Master of Science in Computer Science, GPA: 4.0/4.0 Thesis: <i>Linguistically Differentiating Acts and Recalls of Racial Microaggressions on Social Media</i> [PDF] Research Interests: Natural Language Processing (NLP), Computational Social Science, Human Computer Interaction (HCI), Cultural Analytics Committee: <i>Dr. Eugenia H. Rho (Primary), Dr. Ismini Lourentzou, Dr. Edward Fox</i> ★ Received VT Paul E. Torgersen Research Excellence Award	Blacksburg, VA
May 2021 Aug 2017	Bennett University Bachelor of Technology in Computer Science, CGPA: 9.02/10.00	Greater Noida, India

Experience

Present July 2023	IBM Watsonx Orchestrate Conversational AI powered drive-thrus [🌐] <i>Machine Learning Engineer</i> - Designed and deployed end-to-end spoken language understanding systems (ASR, NLU) optimized for noisy Quick Service Restaurant (QSR) drive-thru environments, operating on resource-constrained edge devices. Served over 80 million clients with the AI-driven dialog system "Archy" [Press]. - Extended real-time multilingual capabilities for Archy by leveraging LLMs to enable Spanish and code-switched (English-Spanish) interactions while maintaining a model size under 7 million parameters. - Implemented optimized training data loaders using bucket sampling, significantly improving training time efficiency and reducing computational overhead by 50% in our PyTorch Lightning-based ASR training pipeline. - Developed a distillation pipeline using multiple large language models (LLMs) to train an edge-deployable NLU and intent detection model tailored for drive-thru orders. - Led the integration of a StyleTTS2 streaming engine in PyTorch into the Watson TTS pipeline, enabling style control and more expressive speech for drive-thru agents.	San Jose, CA
Present July 2023	IBM Research Responsible Tech Team [🌐] <i>Research Collaborator Advisors: Dr. Heloisa Candello (Primary), Dr. Raya Horesch, Dr. Muneeza Azmat</i> - Created a pipeline for aligning large language models (LLMs) to social values, automating synthetic data generation with topic-guided red-teaming and iterative feedback. - Led human evaluation experiments with 60 IBM participants to understand their perceptions of different types of biases in LLMs, using qualitative and quantitative analysis.	San Jose, CA
Aug 2022 May 2022	IBM Research Foundations of AI - Knowledge Reasoning Team [🌐] <i>AI Intern Advisors: Dr. Tengfei Ma (Primary), Dr. Achille Fokoue</i> Developed an interpretable neuro-symbolic model for time series classification, combining neural networks with signal temporal logics to uncover and describe data patterns in human-readable formulas.	Yorktown Heights, NY
Aug 2023 Aug 2021	VT Center for Human Computer Interaction (CHCI) [🌐] <i>Graduate Student Researcher at the Society + AI & Language (SAIL) Lab Advisor: Dr. Eugenia H. Rho</i> - Led foundational mixed-methods HCI research to characterize the language underlying acts and recalls of racial microaggressions in the context of content moderation systems on Reddit. - Leveraged linguistics and NLP to help users counter gray areas of online toxicity. - Used causal inference and prompt-based LLMs to examine the relationship between social media language patterns and trends in national health outcomes.	Blacksburg, VA
Dec 2021 Apr 2020	Indraprastha Institute of Information Technology (IIIT), Delhi MIDAS Lab [🌐] <i>Research Intern (Bachelor Thesis) Advisors: Dr. Debanjan Mahata (Primary), Dr. Rajiv Ratn Shah</i> Developed a framework for abstractive summarization of open-domain Hindi-English code-switched conversations.	New Delhi, India

Aug 2019	Georgia Tech Visual Intelligence Lab	Atlanta, GA
May 2019	Visiting Research Student Mentor: Dr. Devi Parikh	
	> Replicated experiments from a research paper on the behavior of with-attention and without-attention Visual Question Answering (VQA) models.	
Aug 2018	Georgia Tech Chu Data Lab	Atlanta, GA
May 2018	Visiting Research Student Mentor: Dr. Xu Chu	
	> Created a framework for composing and debugging machine learning model workflows using local interpretable model-agnostic explanations (LIME) for enhanced model transparency.	

Publications

S=Under Review/arXiv, W=Workshop, C=Conference

- [C.6] **How Do People Challenge Racial Stereotypes Online? Counter-Story Detection Across Reddit Communities** [Coming Soon]
★ **To appear as a poster at TADA'26**
Uma Gunturi, Jimin Mun, Maarten Sap, Maria Antoniak
The 2026 Conference on Empirical Methods in Natural Language Processing [EMNLP'26]
- [C.5] **Counter-Storytelling in Racial Counterpublics: How Community Context Shapes Narrative Strategies on Reddit** [Abstract], [Poster]
Uma Gunturi, Jimin Mun, Maarten Sap, Long Le-Khac, Maria Antoniak
The International Conference for Computational Social Science [IC2S2'26]
- [C.4] **CounterQuill: Investigating the Potential of Human-AI Collaboration in Online Counterspeech Writing** [PDF]
Xiaohan Ding, Kaike Ping, Uma Gunturi, Buse Carik, Sophia Stil, Lance T Wilhelm, Taufiq Daryanto, James Hawdon, Sang Won Lee, Eugenia H. Rho
IEEE Symposium on Visual Languages and Human-Centric Computing [IEEE VL/HCC'25]
- [C.3] **Linguistically Differentiating Acts and Recalls of Racial Microaggressions on Social Media** [PDF], [Poster], [DEMO] ★ **Received Diversity, Equity, and Inclusion (DEI) Recognition Award**
Uma Gunturi, Anisha Kumar, Xiaohan Ding, Eugenia H. Rho
ACM Conference on Computer-Supported Cooperative Work and Social Computing [CSCW'24]
- [C.2] **Predicting Pandemic Health Decisions and Outcomes Through Social Media Language: A Fuzzy-Trace Theory Approach Leveraging Large Language Models** [PDF]
Xiaohan Ding, Buse Carik, Uma Gunturi, Valerie Reyna, Eugenia H. Rho
ACM Conference on Human Factors in Computing Systems [CHI'24]
- [C.1] **GupShup: Summarizing Open-Domain Code-Switched Conversations** [PDF], [🔗]
Laiba Mehnaz*, Debanjan Mahata*, Uma Gunturi*, Amardeep Kumar*, Rakesh Gosangi, Riya Jain, Gauri Gupta, Isabelle Lee, Anish Acharya, Rajiv Ratn Shah (* = Equal Contribution)
The 2021 Conference on Empirical Methods in Natural Language Processing [EMNLP'21]
- [W.1] **ToxVis: Enabling Interpretability of Implicit vs. Explicit Toxicity Detection Models with Interactive Visualization** [PDF], [DEMO]
Uma Gunturi*, Xiaohan Ding*, Eugenia H. Rho (* = Equal Contribution)
The CHI'23 Workshop on Combating Toxicity, Harassment, and Abuse in Online Social Spaces [CHI'23 Workshop]
- [S.1] **Fill in the ___ in Your Language: Generating Low Resource Language Data from Code Switching** [Coming Soon]
Isabelle Lee*, Uma Gunturi*, Laiba Mehnaz, Zachary Jagers, Teruko Mitamura
arXiv preprint [arXiv]

Honours and Awards

Paul E. Torgersen Research Excellence Award, 2023 | Virginia Tech One of the four recipients in the Master's program
Received Diversity, Equity and Inclusion (DEI) Recognition Award | CSCW 2024 Linguistically Differentiating Acts and Recalls of Racial Microaggressions on Social Media

Teaching and Leadership Roles

SAIL NLP Reading Group, VT Organizer

Aug'21 - Present

- > Organized a weekly lab-wide Reading Group focused on research in the areas of Human Computer Interaction, Natural Language Processing. We read recent and classical papers as well as arrange for invited talks in related areas.

Software Design and Data Structures (CS 2114) Graduate Teaching Assistant

Aug'21 - May'23

- > Responsibilities included evaluating labs, and helping students with the coursework and home/lab assignments.

Skills

- > **Languages:** Java, Python, C++, HTML
- > **Frameworks:** Pandas, NumPy, Gensim, Matplotlib, OpenCV, Keras, Tensorflow, PyTorch, TextBlob, NLTK, HuggingFace
- > **Tools:** Git, GCP, Eclipse, Android Studio, Flask, FireBase, Visual Studio, Stanford Parser, Elasticsearch
- > **Relevant Coursework:** Machine Learning, Information Retrieval, Data Analytics, Human AI Interaction, Natural Language Processing, Deep Learning, Discrete Mathematics, Probability and Statistics

Academic Service

Reviewer CHI'25, AIES'24, CSCW'24, EMNLP'24 Industry Track, CHI'22, CSCW'22

Select Research Projects

Modeling Counter Narratives Across Online Communities

Nov'24 - Present

Advisor: [Dr. Maria Antoniak](#)

- > Leading a research initiative to characterize the language underlying counter-narratives across communities on Reddit. [Accepted@EMNLP'26] [Poster@TADA'26] [Poster@IC2S2'26]
- > Developing a rhetorical codebook to analyze dominant narratives and the character arcs that emerge in counter-narratives. Investigating how AI tools can model these arcs to support users in crafting more persuasive counter-narratives and challenging dominant discourse.

Abstractive Summarization of Open-Domain Code-Switched Conversations

Apr'20 - Dec'21

Advisors: [Dr. Debanjan Mahata](#), [Dr. Rajiv Ratn Shah](#)

- > Introduced the task of abstractive summarization for Hindi-English (Hi-En) code-switched conversations leading to developing the first code-switched conversation summarization dataset *GupShup*. [🔗] [Accepted@EMNLP'21]
- > Experimented with state-of-the-art abstractive summarization models (T5, Pegasus, BART, etc.) and conducted an extensive qualitative analysis of the results to provide insight into some of the model shortcomings.

Intertextuality in Linguistic Interactions

Aug'21 - Present

Advisor: [Dr. Eugenia Rho](#)

- > Led foundational mixed-methods HCI research to characterize the language underlying acts and recalls of racial microaggressions in the context of racism in the U.S. providing broader implications to the current challenges in content moderation systems on Reddit platform. [Accepted@CSCW'24]
- > Introduced Role-Based Incremental Coaching (RBIC), a prompt-based LLM approach grounded in fuzzy-trace theory, to extract critical "gists" from COVID-19 related discussions on Reddit and assessed their impact on online engagement and national health trends such as vaccine uptake and hospitalizations. [Accepted@CHI'24]
- > Visualized semantic opportunities and limitations of transformer-generated counter responses against implicit online aggressions through an interactive UI playground for users to learn and craft responses through machine-generated texts. [🔗] [Accepted@CHI'23 Workshop]

References

- > Dr. Maria Antoniak Assistant Professor, University of Colorado Boulder, USA [🔗]
- > Dr. Heloisa Candello Senior Research Scientist, IBM Research, Brazil [🔗]
- > Dr. Eugenia Rho Assistant Professor, Emory University, USA [🔗]
- > Dr. Debanjan Mahata Senior Research Scientist, Bloomberg, USA [🔗]
- > Dr. Muneeza Azmat Research Scientist, IBM Research, USA [🔗]